MBA: 8109 RESEARCH METHODS



Week 7: Population and Sampling

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SESSION ROADMAP

Where Section 3.2 sits, and what you will leave with to night

3.1 Research design	**3.2 Population & sampling**	3.3 Data collection	3.4 Data analysis	3.5 Research quality	3.7 Ethics

SU Research Guide, §7.2(b) — Chapter 3 structure of the proposal

Learning outcomes — by the end of tonight you can:

- Define population, target and accessible population, unit of analysis, frame and sample
- Match the population logic to the kind of data your study needs
- Source and cite a scientific, verifiable population figure
- Show the chain from Chapter 1 scope to the Chapter 3 frame
- Choose between probability and non-probability techniques
- Allocate a sample proportionately or disproportionately across strata, clusters and quotas

Tonight in two parts

01 Defining the population

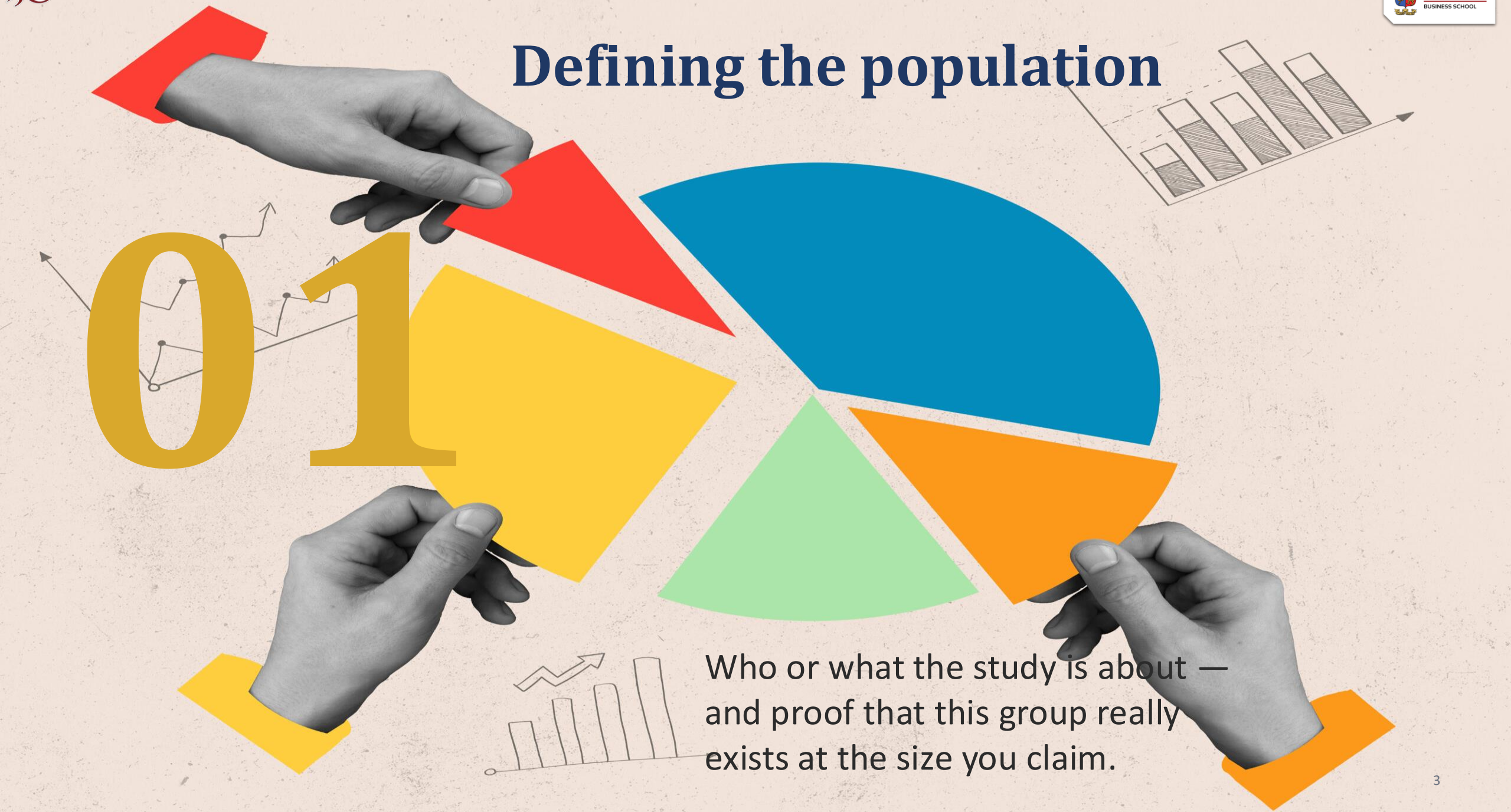
Definitions · data types · verifiable sources · link to scope

02 Sampling frame and techniques

Frame quality · probability vs non-probability · proportionate and disproportionate allocation · sample size · writing 3.2

Defining the population

01



Who or what the study is about — and proof that this group really exists at the size you claim.

KEY TERMS

The vocabulary of Section 3.2

Population

The full set of cases (people, firms, facilities, records, transactions) from which a sample is drawn and about which conclusions are made.

Target population

The population the research question is ultimately about, e.g. all nurses in Kenyan level 4–6 public hospitals.

Accessible population

The part of the target population you can actually reach, e.g. nurses in Nairobi County level 4–6 hospitals.

Unit of analysis

The level at which data are analysed: individual, team, firm, facility, county, year. It must match the objectives.

Sampling frame

The complete, named list of all cases in the accessible population from which the sample is actually selected.

Sample and census

A sample is the subset selected; a census collects data from every case — advisable where the population is small (under about 50).

Source: Saunders, Lewis & Thornhill (9th ed.), §7.1–7.2, pp. 289–293.

Your data type decides how the population is treated

Kind of study / data	How the population is framed	Typical selection	Generalisation claimed
Quantitative primary (survey, experiment)	Countable, listed cases with a known total N	Probability sampling; sample size by formula or table	Statistical — to the population in the frame
Qualitative primary (interviews, FGDs, case study)	Information-rich cases or participants with the relevant experience	Purposive, snowball; size by saturation	Analytical / theoretical — to concepts, not to N
Secondary / archival (financial statements, registers)	Records, firms or firm-years over a defined period	Census of the listed set, or a filtered subset with stated criteria	To the defined set of records and period
Mixed methods	Two populations or two layers (e.g. staff and managers)	Probability for the quantitative strand, purposive for the qualitative strand	Each strand to its own logic; explain how they integrate

Rule: state the data type first, then the population logic — never borrow a survey sample-size formula for an interview study.

Source: Saunders et al. (9th ed.), §7.3 p. 293 and §7.8 p. 313.

Every population figure needs a source an examiner can check

If your population is…	Verifiable source to cite (with year)
Listed companies	Nairobi Securities Exchange listing; Capital Markets Authority reports
Commercial banks, MFBs, SACCOs	Central Bank of Kenya supervision report; SASRA register
Health facilities	Kenya Master Health Facility List (Ministry of Health)
Households, population, SMEs	KNBS Census and Economic Survey; MSME survey
Employees of one organisation	HR establishment or payroll record, with a gatekeeper letter
Professionals	Registers of ICPAK, LSK, KMPDC, NCK, EBK



What makes it "scientific"

Official or institutional, dated, publicly or formally obtainable, and consistent with the scope. Put the list in Appendix 3 of the proposal, as the SU Guide provides.

Red flags examiners spot

"About 500 employees" with no source

A figure from a news story or website

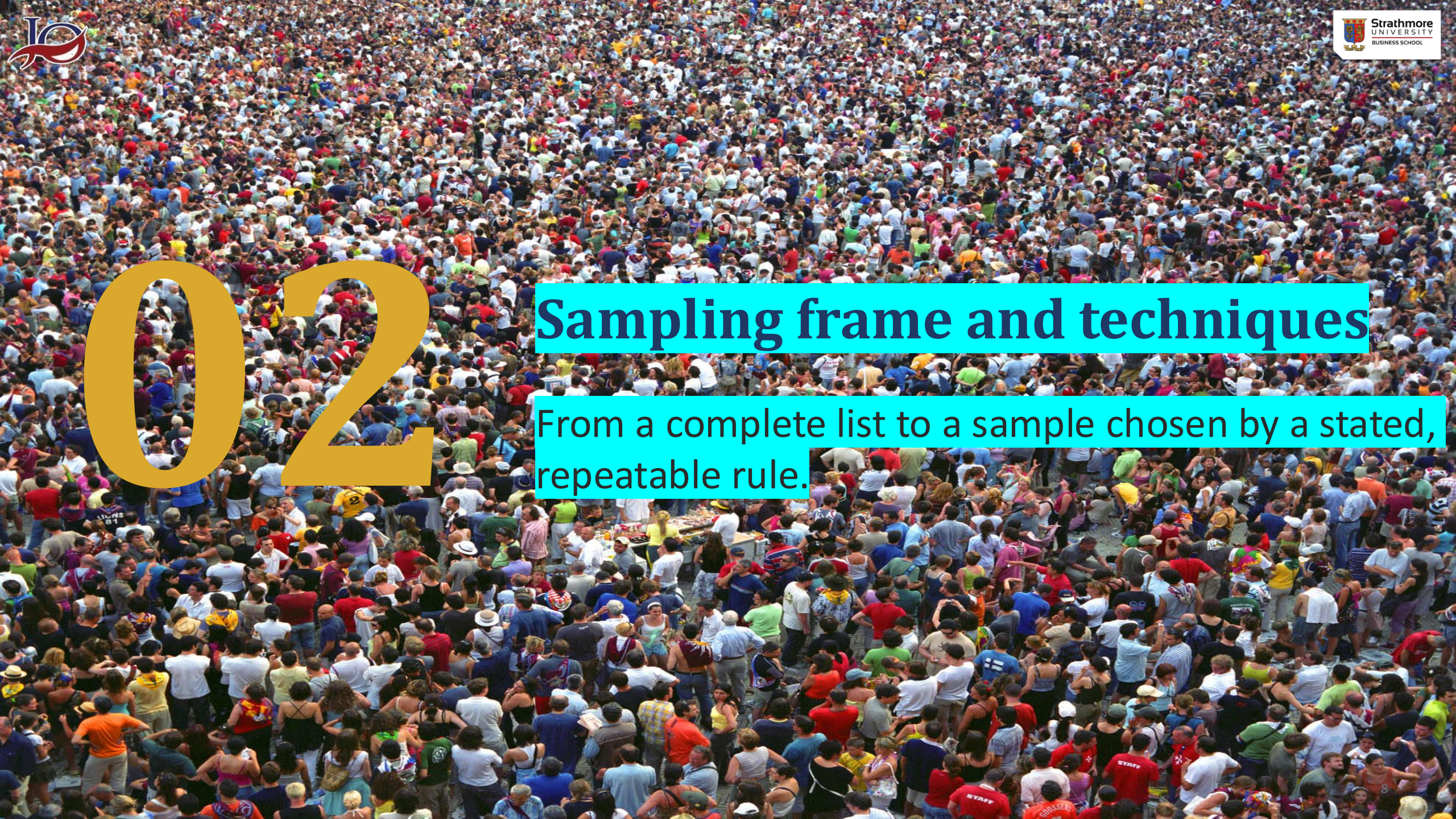
A list older than the study period

Numbers that do not add up across strata

Chapter 1 scope (1.5) sets the boundaries that 3.2 must honour

Chapter 1 · Scope		Chapter 3 · Population and sampling
Content scope Variables and constructs studied	→	Who holds the information → respondents and unit of analysis
Geographic scope County, sector, region, country	→	Which cases enter the frame
Time scope Period of study, e.g. 2019–2025	→	Firm-years or a cross-section date
Methodological scope Design and data type	→	Probability vs non-probability

Consistency test: title → objectives → scope → population → frame → sample must describe the same cases. You may generalise only to the frame you sampled from.



02

Sampling frame and techniques

From a complete list to a sample chosen by a stated, repeatable rule.

A good frame is relevant, current, complete, accurate and precise

1. **Relevant**
   Cases can answer the research questions

2. **Up to date**
   Compiled recently, matching the time scope

3. **Complete**
   Every eligible case is listed

4. **Accurate**
   Names, sizes and categories are correct

5. **Precise**
   Irrelevant cases are excluded

Common frame errors

Under-coverage: eligible cases missing, e.g. informal traders

Over-coverage: closed or ineligible cases still listed

Duplication: branches listed as separate firms

Clustering: one entry covering many cases

How to present it in 3.2 — population distribution table

Category	Population (N)	Source
Stratum / cluster A	…	Register, year
Stratum / cluster B	…	Register, year
Total	N	

Source: Saunders et al. (9th ed.), §7.4 pp. 295–297, checklist for selecting a sampling frame.

Probability versus non-probability sampling

	Probability (representative)	Non-probability (judgemental)
Basis of selection	Random; each case has a known, usually equal, chance	Researcher judgement or access; chance unknown
Sampling frame	Required	Not always required
Generalisation	Statistical inference to the population	Analytical, to theory or the case
Sample size	Calculated from confidence level and margin of error	Set by purpose, saturation, feasibility
Typical designs	Surveys, experiments, correlational studies	Case studies, interviews, exploratory and pilot work
Techniques	Simple random, systematic, stratified, cluster, multi-stage	Quota, purposive, snowball, self-selection, convenience

Five ways to select randomly, and when each fits

Simple random

Every case numbered; selected by random numbers.

Use: small, listed, not dispersed populations.

Systematic

Every k-th case after a random start (k = N / n).

Watch: periodic patterns in the list.

Stratified random

Divide the frame into strata (sector, grade, size), then sample randomly within each.

Use: to guarantee sub-group representation.

Cluster

Randomly pick whole natural groups (branches, counties, schools) and study all cases in them.

Trade-off: cheaper, less precise.

Multi-stage

Clusters, then sub-clusters, then cases, random at each stage (e.g. county → facility → staff).

Needs a frame only at the final stage.

Mixed designs

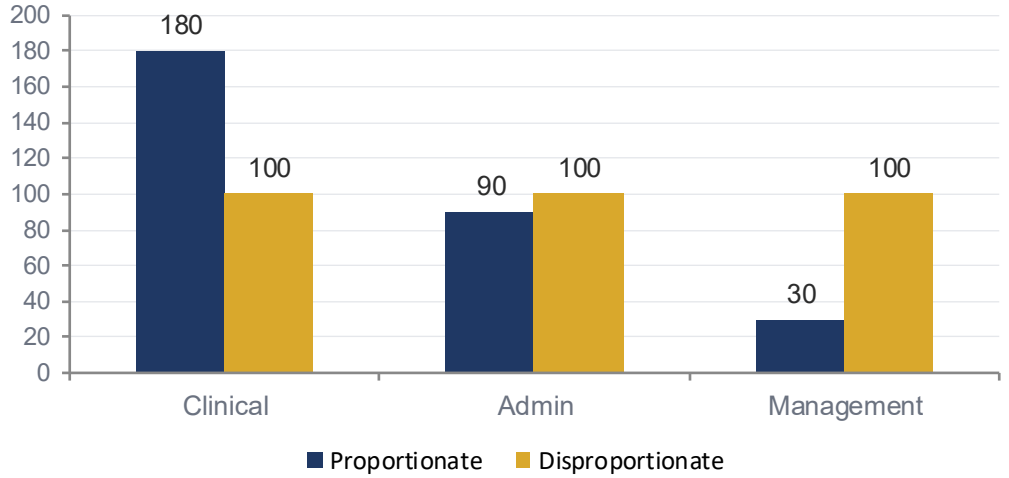
Most MBA studies combine techniques, e.g. stratify by department, then select systematically within each stratum. Describe each stage in 3.2.

Proportionate vs disproportionate allocation

Worked example: 1,200 staff in three strata, n = 300

Stratum	N	Share	Proportionate	Disproportionate (equal)	Weight N/n
Clinical	720	60%	180	100	7.2
Administrative	360	30%	90	100	3.6
Management	120	10%	30	100	1.2
Total	1,200	100%	300	300	—

Sample per stratum



Proportionate: n(h) = n × N(h) / N

- Each stratum mirrors its share of the population; results can be pooled directly
- Disproportionate: over-sample small but important strata so each has enough cases to analyse
- Disproportionate results must be weighted back (N/n) before any population-wide estimate

Clusters and multi-stage

Proportionate: select clusters with probability proportional to size (PPS), so large branches are more likely to enter.

Disproportionate: equal chance per cluster regardless of size — weight by cluster size at analysis.

Source: Saunders et al. (9th ed.), §7.6 pp. 304–312 on stratified, cluster and multi-stage sampling and weighting strata.

Judgement-based selection — and proportion still matters

Technique	How cases are chosen	Best used when
Quota — proportional	Quotas mirror population shares (e.g. 60% clinical, 30% admin, 10% management)	No frame, but known population proportions
Quota — non-proportional	Fixed minimum per group regardless of size	Sub-group comparison matters more than mirroring
Purposive	Typical, extreme, critical, homogeneous or maximum-variation (heterogeneous) cases	Information-rich cases for depth
Snowball	Initial participants refer others	Hard-to-reach or hidden populations
Self-selection	Cases volunteer in response to a call	Topic draws motivated participants
Convenience	Whoever is easiest to reach	Pilots only; weakest justification

Proportion without randomness

Proportional quota is the non-probability twin of proportionate stratified sampling; non-proportional quota mirrors disproportionate allocation.

In purposive designs

Balance across strata comes from selection criteria (e.g. two managers per department). State the inclusion and exclusion criteria.

Justify the number, by the logic of the technique

Probability samples

Yamane (1967): $n = N / (1 + N e^2)$

N = 1,200, e = 0.05 → n ≈ 300

Also accepted: Krejcie & Morgan (1970) table; Cochran (1977) formula for large populations

Adjust for non-response: n(adj) = n × 100 / re%

n = 300, expected response 75% → issue 400

State confidence level (usually 95%) and margin of error; populations under about 50 → census

Non-probability samples

No formula — the logic of purpose governs

Saturation: stop when new interviews add few new insights

Homogeneous group: about 12 in-depth interviews (Guest et al., 2006)

Heterogeneous or broad question: about 25–30 interviews

Quota: large enough per quota cell for the planned analysis

Justify with purpose, depth, resources and access — not with a survey formula

Writing Section 3.2 — and tonight's practical

3.2.1 Target population
Define it, give N and cite the verifiable source; population table

3.2.2 Sampling frame
Name the list, its date, and inclusion or exclusion criteria

3.2.3 Sample size
Formula or logic, parameters and the calculation shown

3.2.4 Technique and procedure
Probability or not; proportionate or disproportionate allocation; step by step

Sample distribution table
Stratum or cluster · N · n · technique within each

Practical

For your own concept paper:

1. Quote your Chapter 1 scope in one line
2. State the target population with its source and year
3. Name your frame and one likely frame error
4. Choose probability or non-probability, and say why
5. Draft the sample distribution table, proportionate or disproportionate

Pairs swap and apply the consistency test from slide 7.



Reading ahead: Saunders et al. (9th ed.), Ch. 7 self-check questions and "Progressing your research project", pp. 331–332.